



General histology

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- Polarity
- Classification of epithelial tissue.
- Simple epithelium.
- Types
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- Types

Objectives:

- Definition of polarity
- Student must understand different types of simple and stratified epithelial tissues.
- Student must understand structure of each type of the epithelial tissues.

Polarity:

- The stable and organized structural and functional differences within a cell where different regions of the cells are specialized to perform different function.
- The cell is not the same on all sides, different parts of cells have different structures and functions.

Polarity in epithelial cells means:

The cell has three distinct surfaces:

- Apical surface → faces the lumen or outside, involved in absorption or secretion.
- Lateral surface → contacts neighboring cells, involved in adhesion and communication
- Basal surface → attaches to the basement membrane

Classification of epithelial tissues

- 1- shape of the cells at the free surface .
- 2- number of cell layers of the epithelial membrane.
- 3-Function.

1- shape of the cells at the free surface .

a. Squamous epithelium

- Cells are flat and thin
- Nucleus is flattened
- Function: diffusion, filtration
- Examples: alveoli of lungs, blood vessels

1- shape of the cells at the free surface .

b. Cuboidal epithelium

- Cells are cube-shaped
- Nucleus is round and central
- Function: secretion and absorption
- Examples: kidney tubules, glands

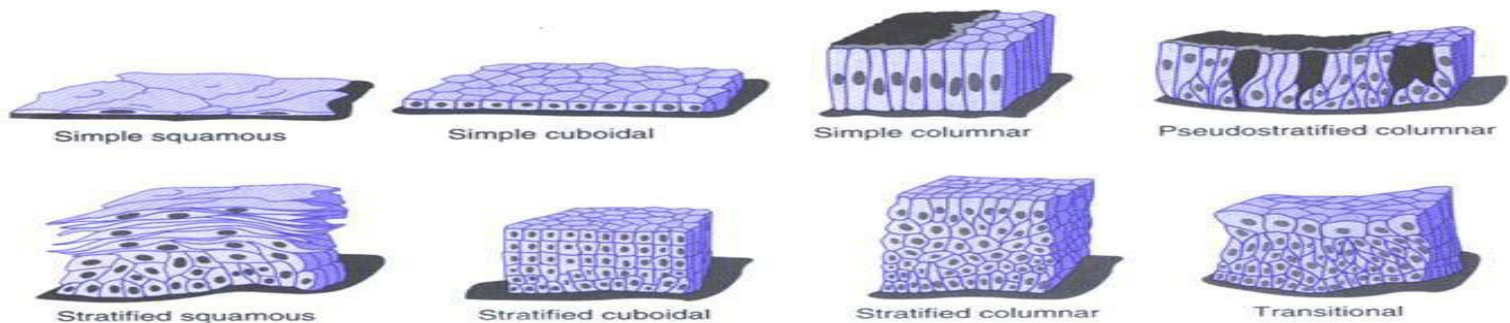
1- shape of the cells at the free surface .

c. Columnar epithelium

- Cells are tall and column-shaped
- Nucleus is elongated and basal
- Function: absorption and secretion
- Examples: stomach, intestines

2- number of cell layers of the epithelial membrane.

- Simple: one layer of cells.
- Stratified: more than one layers of cells



Simple epithelium

- Consists of a single layer of cells
- All cells rest on the basement membrane
- Functions: absorption, secretion, diffusion, filtration
- Examples:
 - Simple squamous → alveoli, blood vessels
 - Simple cuboidal → kidney tubules
 - Simple columnar → intestine

Stratified epithelium

- Consists of two or more layers of cells
- Only the basal layer rests on the basement membrane
- Main function: protection
- Named according to the shape of the surface (apical) cells
- Examples:
 - Stratified squamous → skin, oral cavity
 - Stratified cuboidal → sweat glands
 - Stratified columnar → rare, large ducts

Special types:

- **Pseudostratified epithelium**
- Appears stratified but is actually one layer
- All cells touch the basement membrane
- Example: respiratory tract
- **Transitional epithelium**
- Specialized stratified epithelium
- Allows stretching
- Example: urinary bladder.

3. Function of epithelium :

- Covering and lining epithelium
- Covers the body surface
- Lines body cavities, organs, and ducts

Main functions:

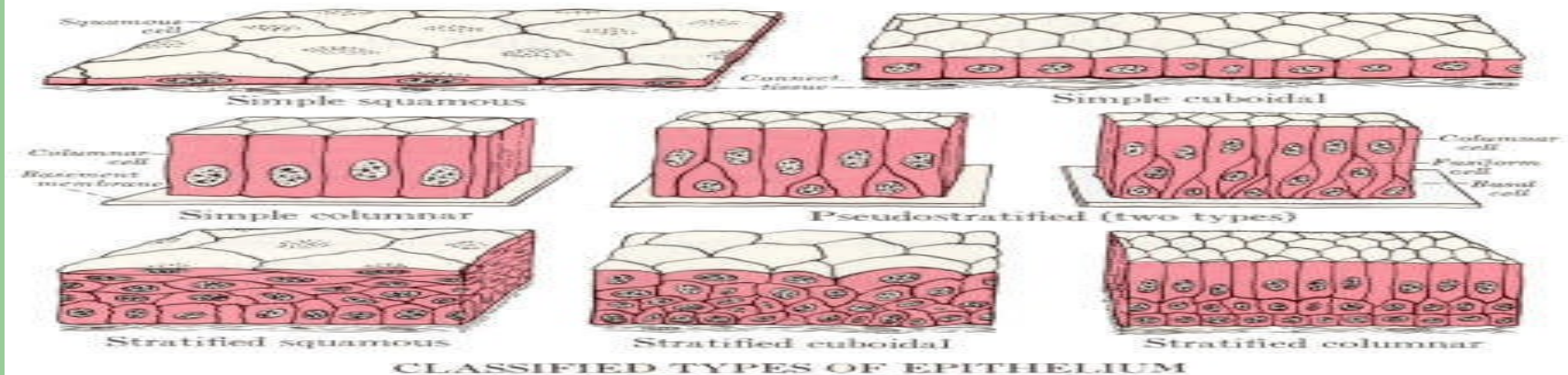
Protection, Absorption, Secretion, Filtration

Examples:

Epidermis of skin, Lining of gastrointestinal tract, Blood vessels

A- Simple epithelium:

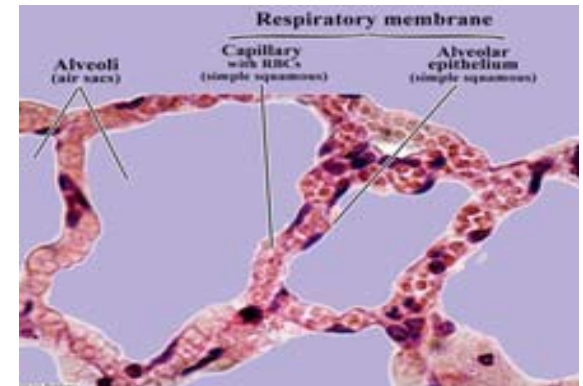
- Simple epithelia consist of a single layer of cells, and all cells rest on the basement membrane.

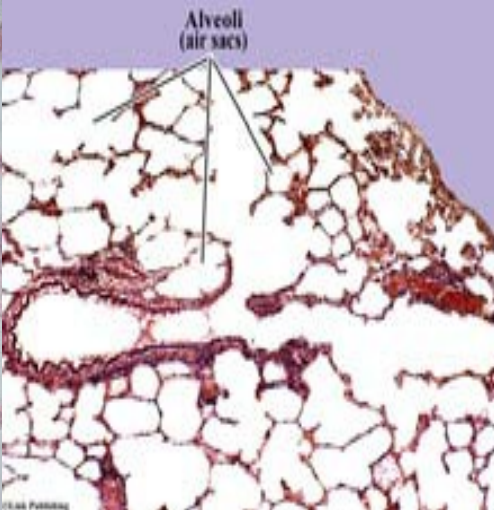
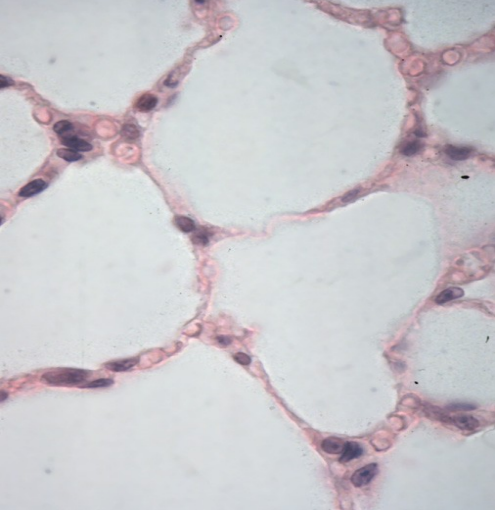


Simple epithelia can be divided according to the morphology of the cells into:

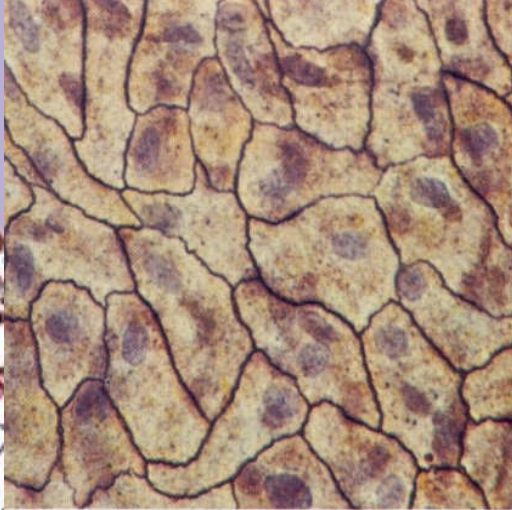
1- simple squamous epithelium.

- Single layer of flattened cells with disc shaped nuclei and scant cytoplasm.
- **-Location:** lining Bowman's space in the cortex of the kidney. lining body cavity (mesothelium), cardiovascular and lymphatic systems, surface of lung and kidneys (gas and liquid exchange) lines blood vessels (endothelium)
- **function =** diffusion and filtration.





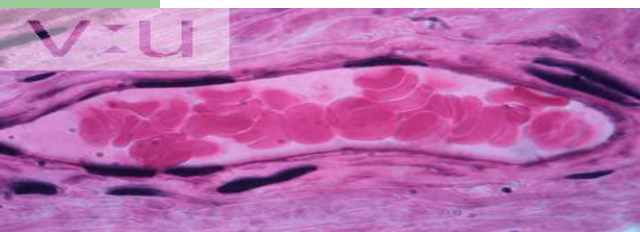
Lung alveoli



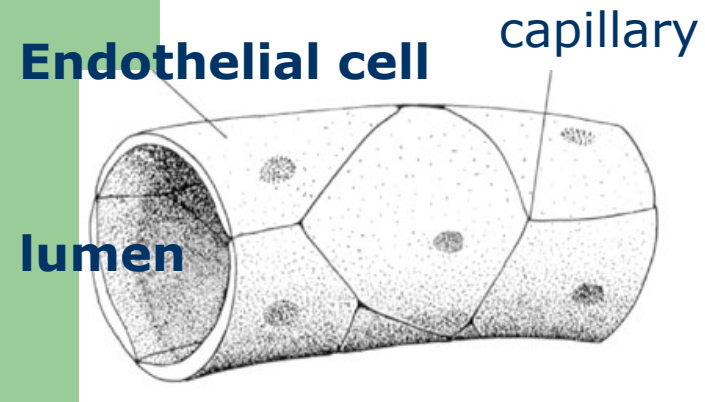
Mesothelium - Silver (Oil)



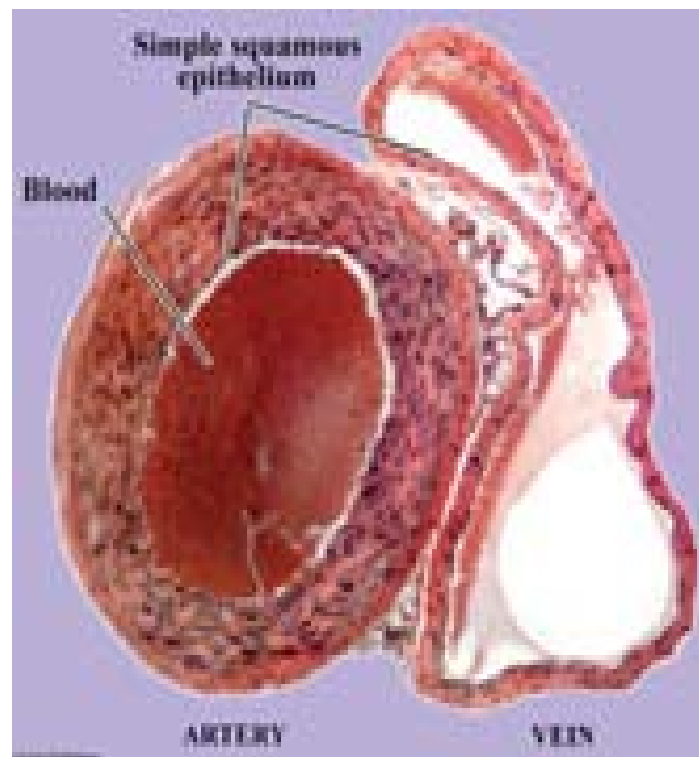
peritoneum



Endothelial cell
(simple squamous)



Blood capillary



2-Simple cuboidal epithelium

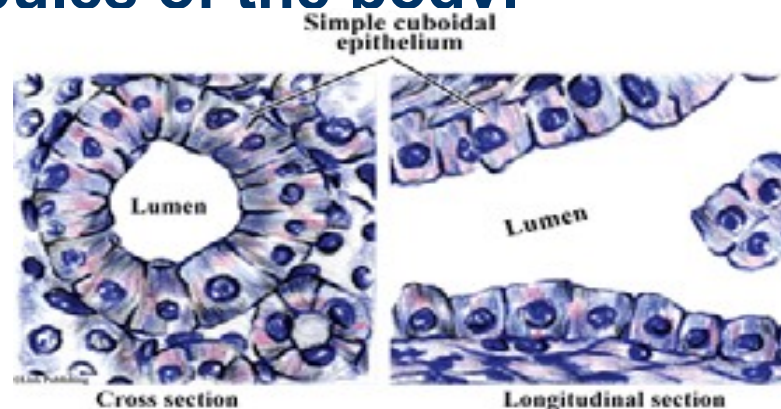
Is formed by a single layer of cuboidal cells with round centrally located nucleus.

• Locations

- 1-lines most of the kidney tubules.
- 2-lines the excretory duct of glands and secretory portion of many glands.
- 3- lining many ducts and tubules of the body.
- 4-covers the ovary.

• Functions

- secretion and absorption



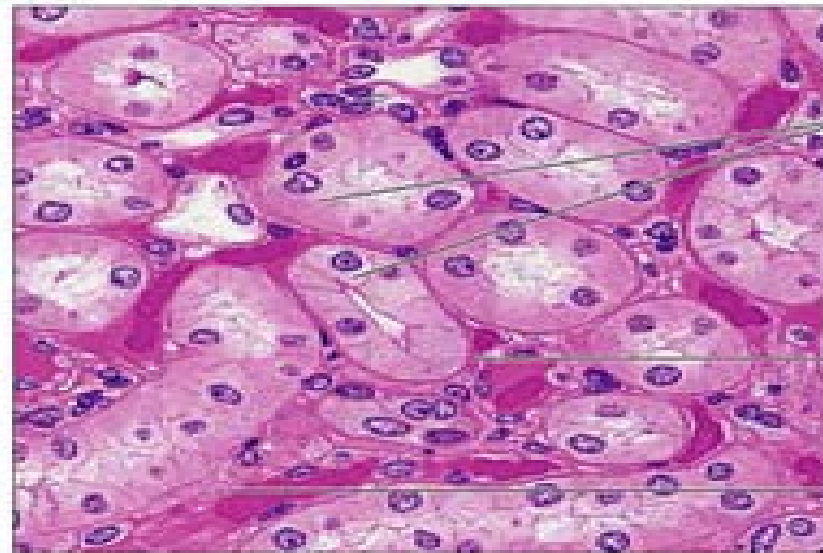
(b) Simple cuboidal epithelium

Description: Single layer of cubelike cells with large, spherical central nuclei.



Function: Secretion and absorption.

Location: Kidney tubules; ducts and secretory portions of small glands; ovary surface.



Simple cuboidal epithelial cells

Basement membrane

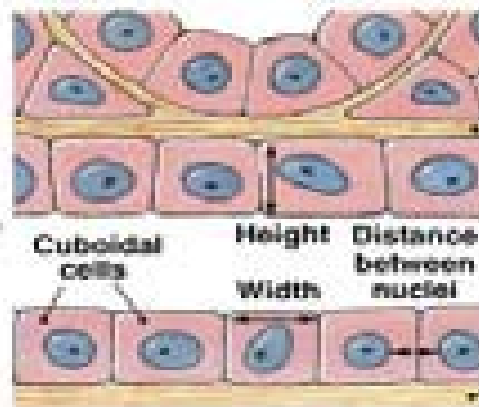
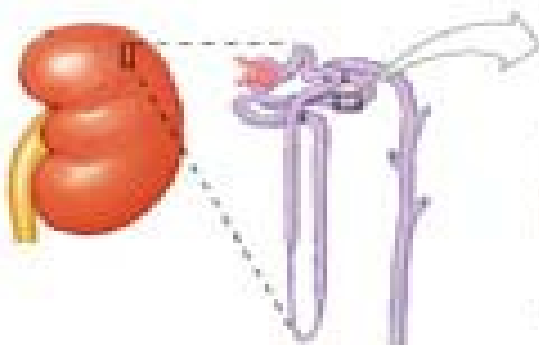
Connective tissue

Photomicrograph: Simple cuboidal epithelium in kidney tubules (400 $\times$).

SIMPLE CUBOIDAL EPITHELIUM

LOCATIONS: Glands; ducts; portions of kidney tubules; thyroid gland

FUNCTIONS: Limited protection, secretion, absorption



Nucleus

Cuboidal cells

Height
Width

Distance between nuclei

Connective tissue

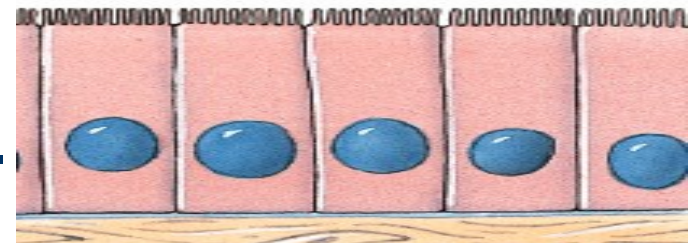
Basal lamina



(a) Kidney tubule

3-Simple columnar epithelium

- Composed of a single layer of cells that are tall, column-shaped cells. Cells are taller than they are wide. Elongated nuclei are located near the basal part of cell. Nuclei usually appear in a single row in cross section.
- **Locations:**
 - -found lining the stomach, intestines, and some large ducts in the body. often have surface modifications such as cilia and microvilli
- **function** secretion and absorption.



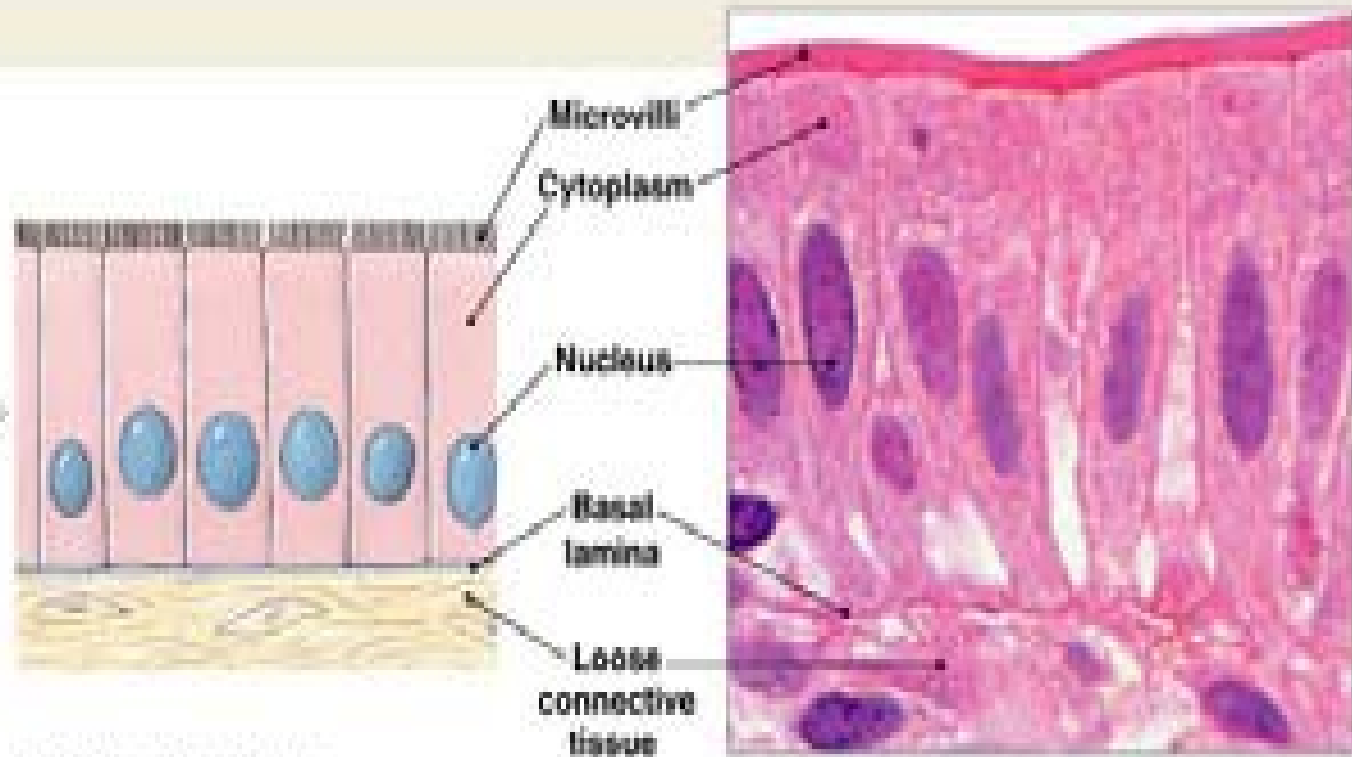




SIMPLE COLUMNAR EPITHELIUM

LOCATIONS: Lining of stomach, intestine, gallbladder, uterine tubes, and collecting ducts of kidneys

FUNCTIONS: Protection, secretion, absorption



(a) Intestinal lining

LM $\times 350$

4-Pseudostratified Columnar epithelium

Single layer of cells with different heights; all cells rest on basement membrane, but some do not reach the free surface, Nuclei are seen at different levels.

- **Locations**

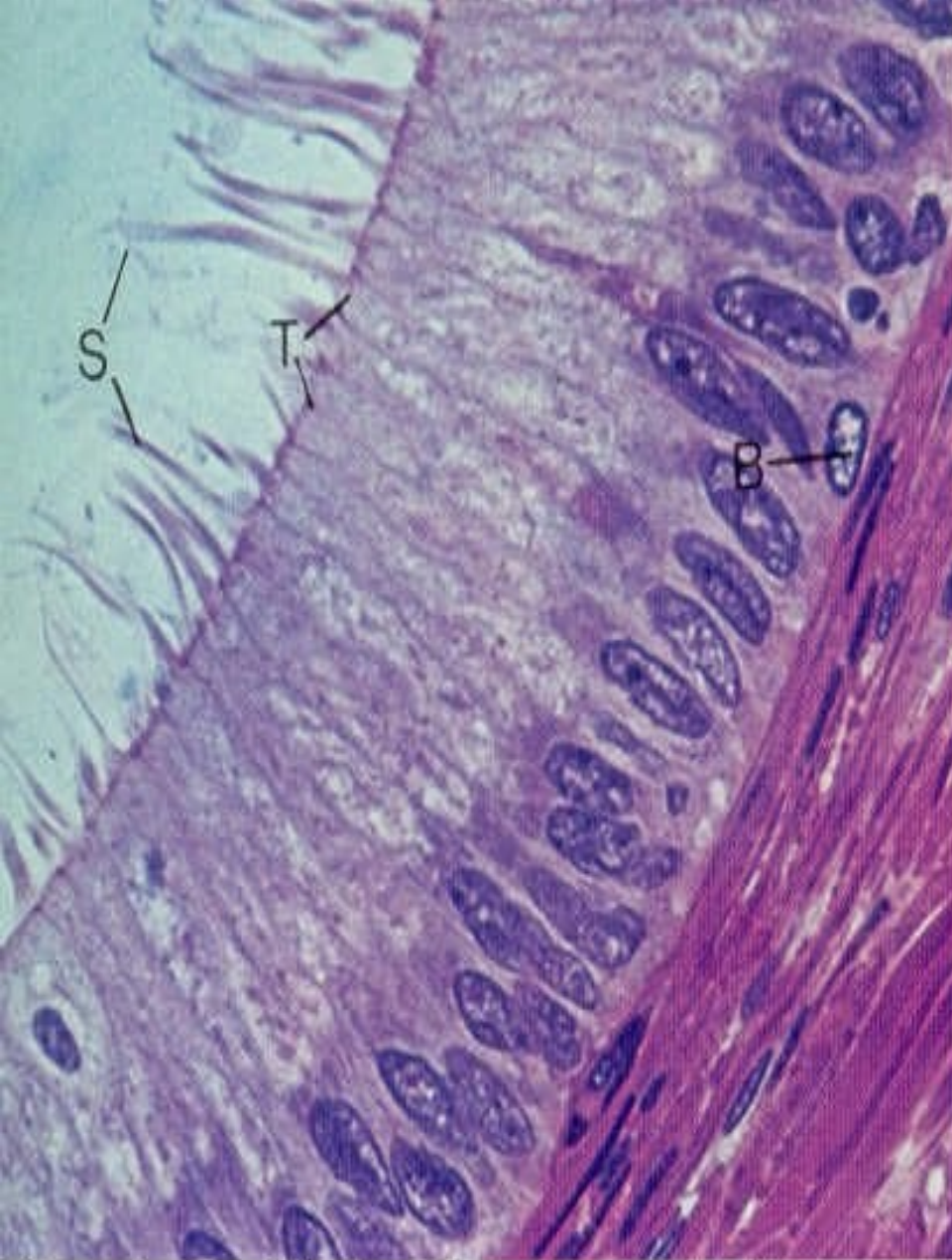
1-lines most of the nasal cavity, trachea, bronchi.

2-lines some of the male reproductive tract (epididymis).

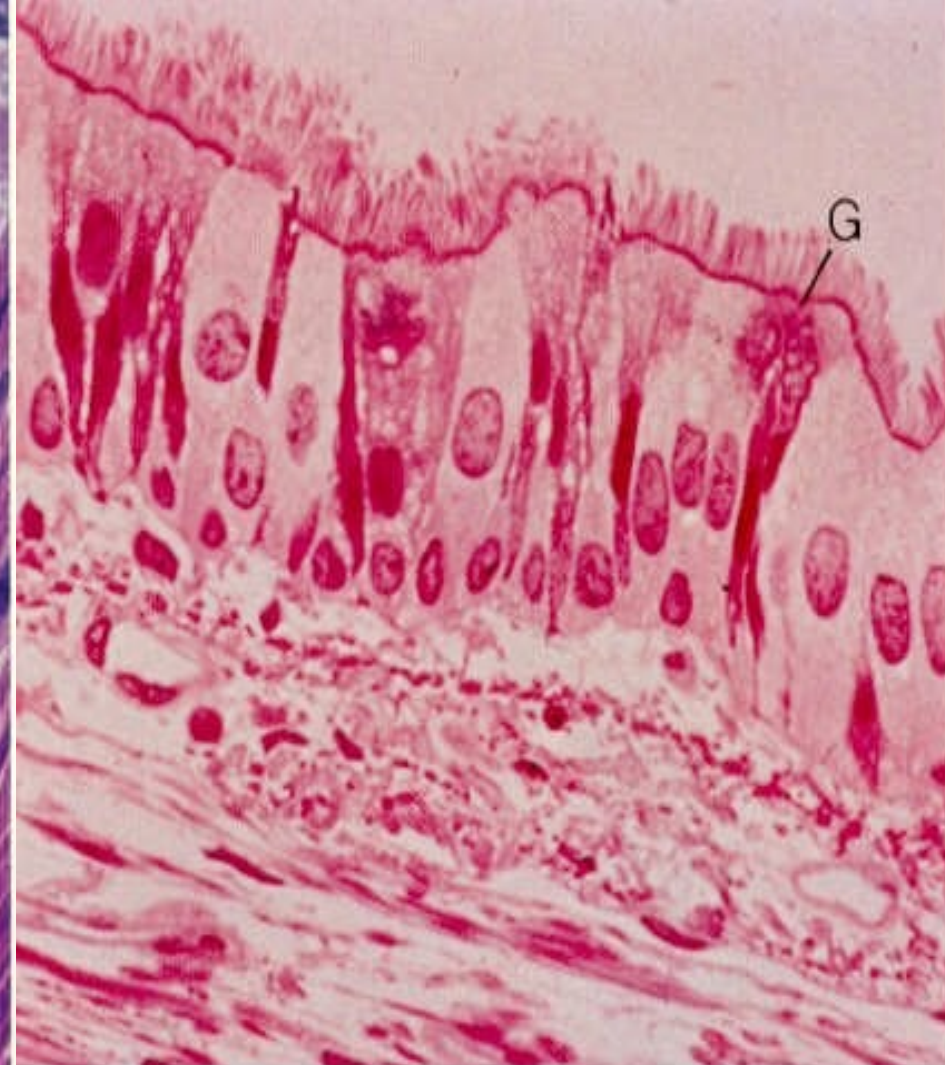
- **Functions**

- protection, secretion, and the movement of substances (mucus) over the surface





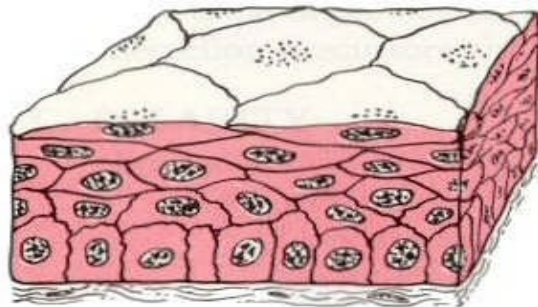
Pseudostratified columnar epithelium
Ductus Epididymis H&E (High X)



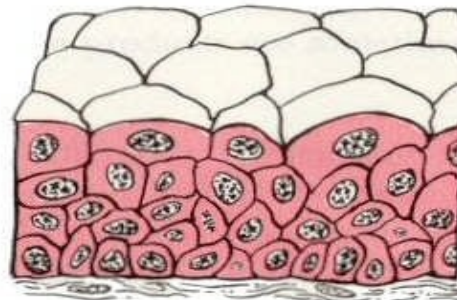
Trachea, transverse sec. H&E (Med)
(G) goblet cells in Pseudostratified
ciliated columnar epithelium. Lamina
propria is loose areolar C.T.

Stratified epithelia:

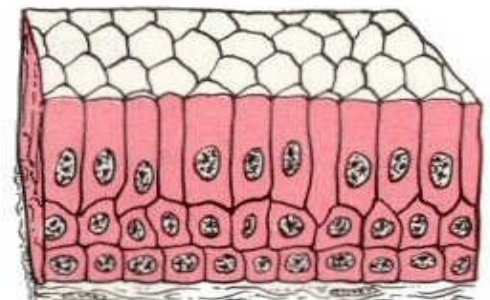
- Stratified epithelia consists of two or more cell layers.
- Subtypes are named according to the shape of the surface (apical) cells alone.



Stratified squamous



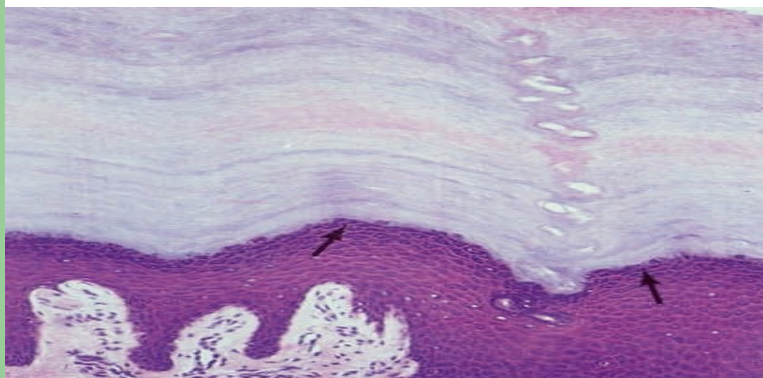
Stratified cuboidal



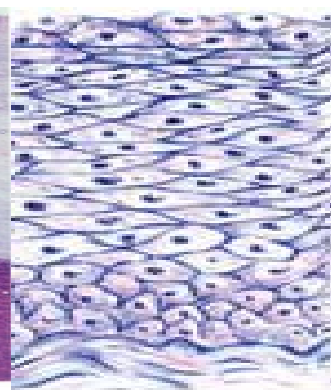
Stratified columnar

1-Stratified Squamous Epithelium

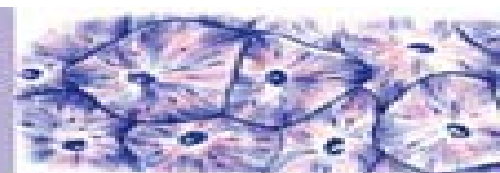
- Is composed of many layers of cells with squamous cells on the surface. The middle are cuboidal while the basal cells that resting on a basal lamina are columnar, this types occurs in two forms are :



Thick skin of sole of foot H&E (Med)
(arrows) = stratum granulosum



Stratified squamous epithelium



Squamous surface cells



Mitotic basal cells

Types of stratified squamous epithelium:

- **1. Keratinized stratified squamous epithelium:**
- Surface cells are dead and filled with keratin
- Provides strong protection and prevents water loss
- Example:
- Epidermis of the skin

Types of stratified squamous epithelium:

- **2. Non-keratinized stratified squamous epithelium**
- Surface cells are living and moist
- Protects against abrasion in wet surfaces
- **Examples:**
- Oral cavity
- Esophagus
- Vagina

Functions of Stratified Squamous Epithelium:

- **Primary function:**
- Protection of underlying tissues from:
- Mechanical abrasion
- Physical stress
- Pathogens.

Functions of Stratified Squamous Epithelium:

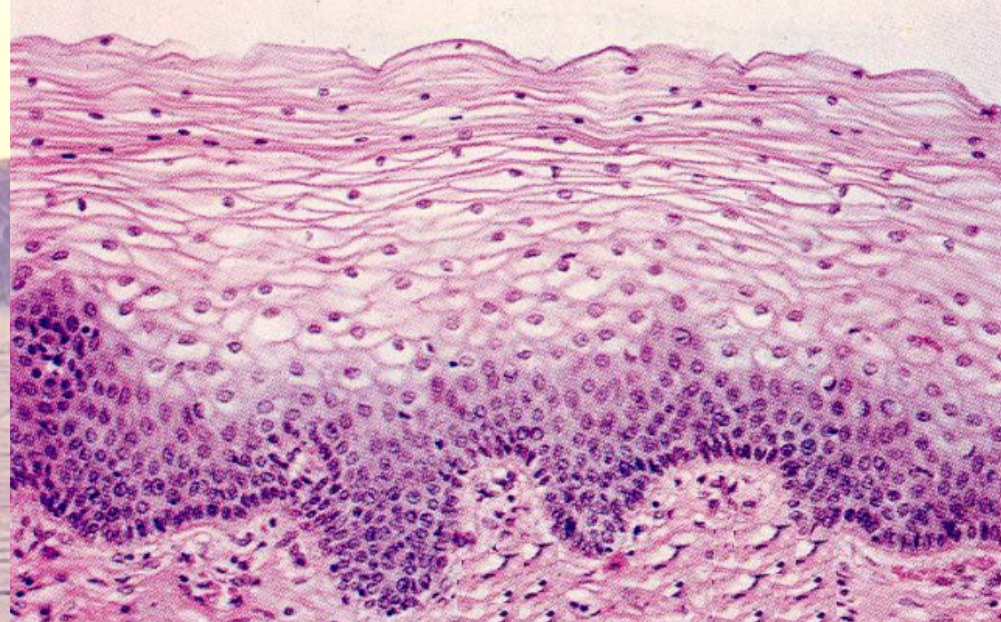
- **Keratinized stratified squamous epithelium (epidermis):**
 - Forms a tough, dry barrier
 - Prevents water loss
 - Protects against chemical and microbial damage •

Functions of Stratified Squamous Epithelium:

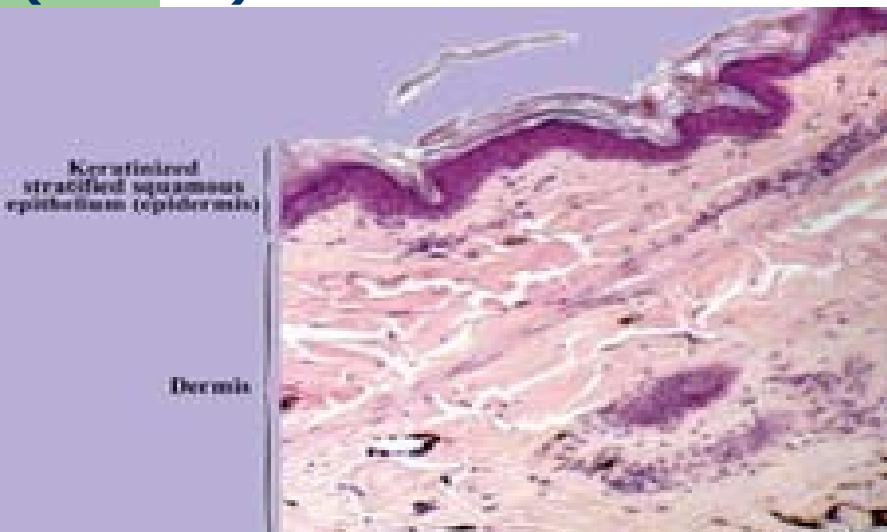
- **Non-keratinized stratified squamous epithelium:**
- Protects moist surfaces
- Resists friction and abrasion
- Found in oral cavity, esophagus, vagina



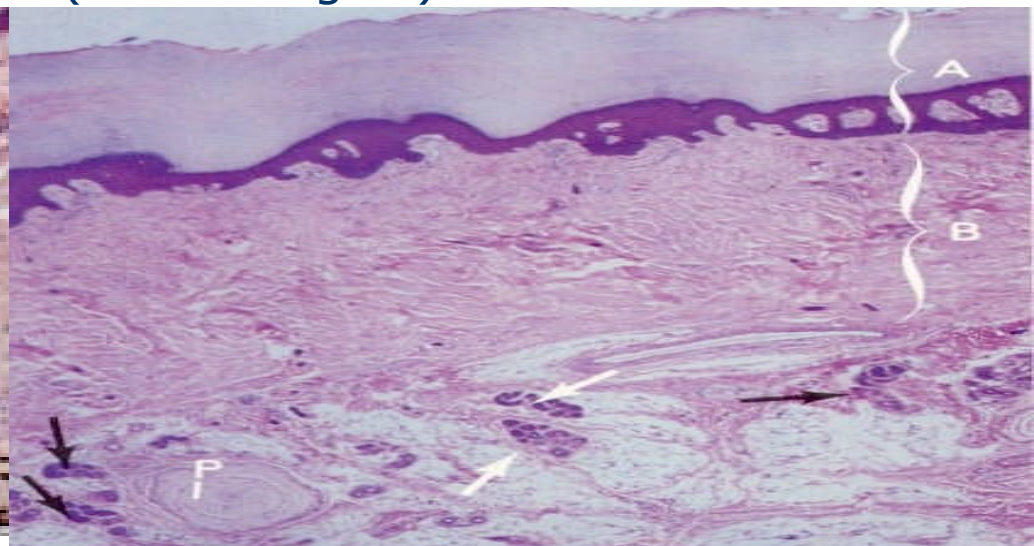
Stratified squamous epithelium (cornea)



Stratified squamous epithelium (wall of vagina)



1-Stratified Squamous Keratinized epithelium (epidermis)



Thick skin of sole of foot H&E (Low)
 (A) = epidermis; (B) = dermis
 (arrows) = sweat glands;
 (P) = Vater-Pacini corpuscle

2-Stratified Cuboidal Epithelium:

Quite rare in the body. Typically 2 to 3 cell layers thick only top layer is cuboidal.

Locations

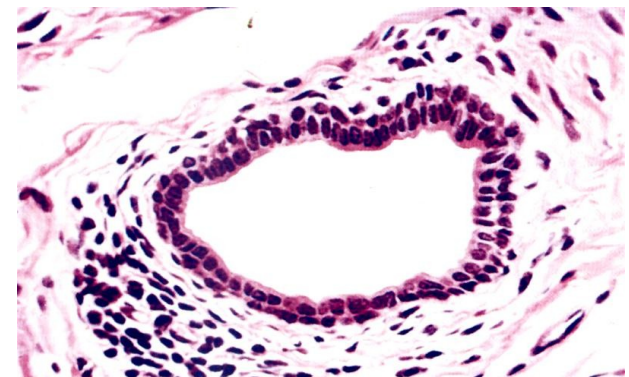
Testis tubules; vesicular (Graafian) follicles of ovary.

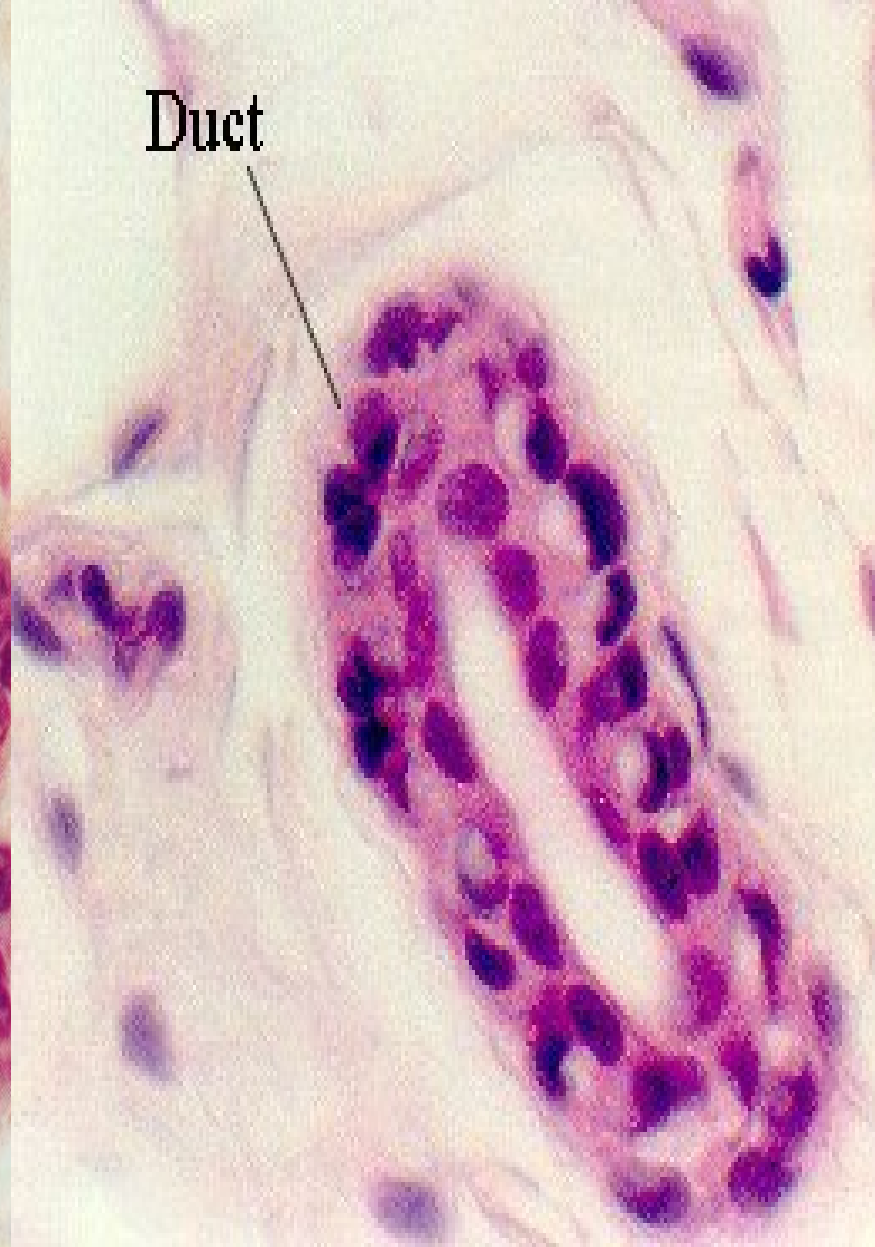
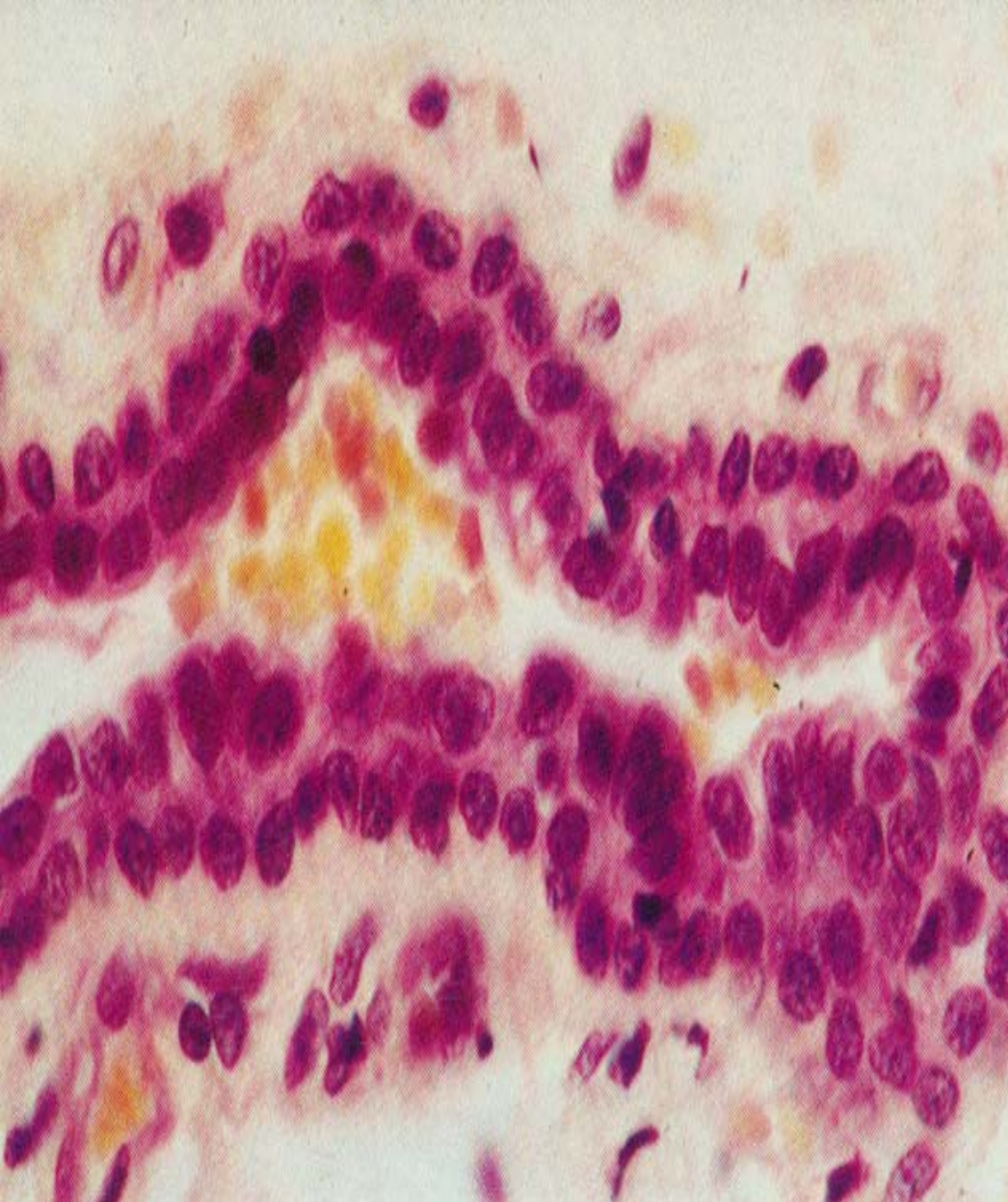
Ducts of sweat glands; sebaceous glands.

Intermediate zones of some regions, such as the urethra and conjunctiva.

- **Functions**

Protection, limited Secretion and absorption





Stratified cuboidal epithelium (lactiferous duct of breast)	Ducts of sweat glands
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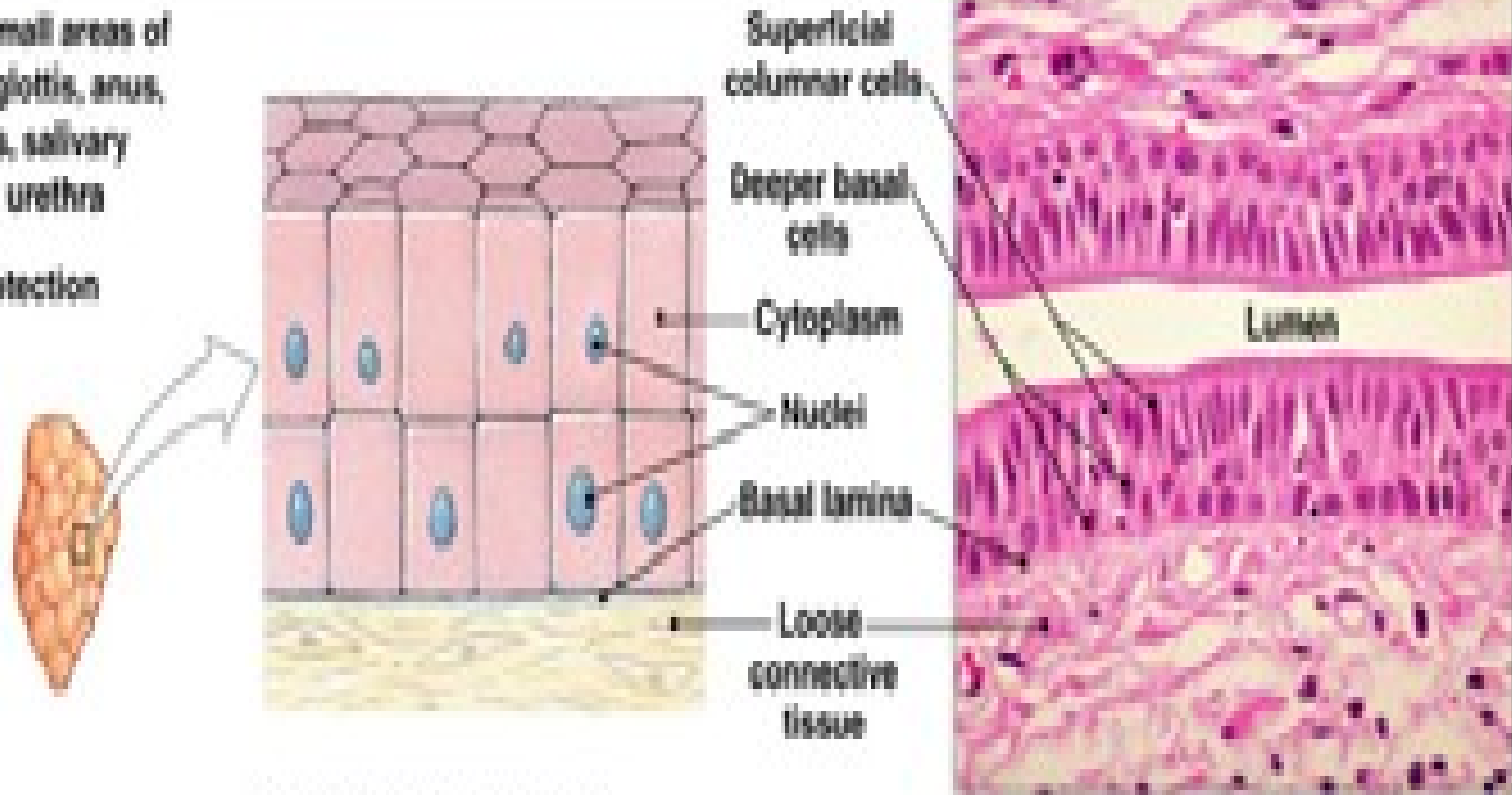
3-Stratified Columnar Epithelia:

- **Definition**
- A rare type of stratified epithelium
- Consists of two or more layers of cells
- Superficial (apical) cells are columnar
- Basal cells are usually cuboidal
- May be ciliated in some locations

STRATIFIED COLUMNAR EPITHELIUM

LOCATIONS: Small areas of the pharynx, epiglottis, anus, mammary glands, salivary gland ducts, and urethra

FUNCTION: Protection



(c) Salivary gland duct

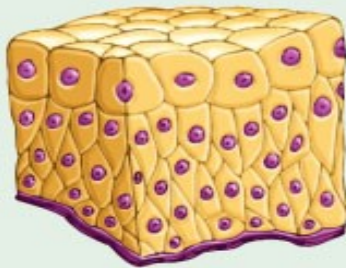
4-Transitional Epithelia

- **Definition**
- A specialized stratified epithelium
- Composed of several layers of cells
- Basal cells are cuboidal or columnar
- Surface (umbrella) cells are large, dome-shaped when relaxed and may be binucleated
- Surface cells become flattened when stretched

4-Transitional Epithelia

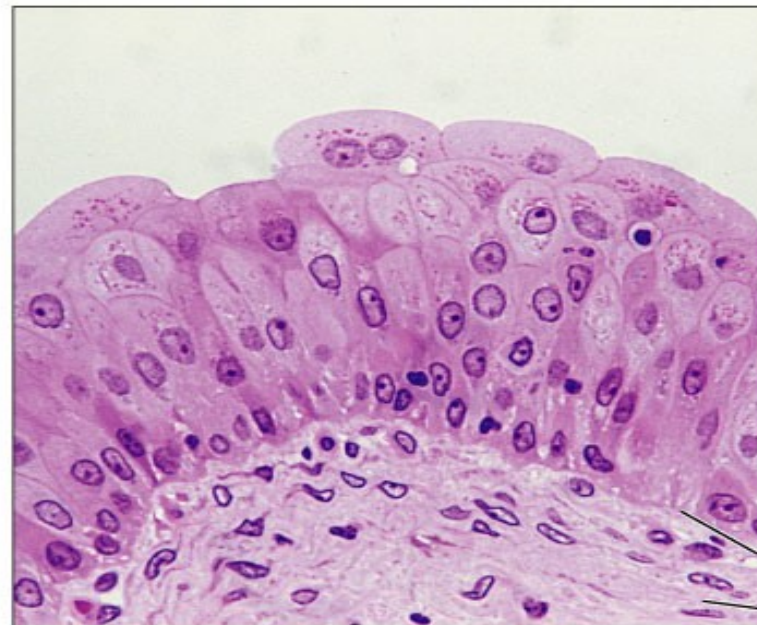
(f) Transitional epithelium

Description: Resembles both stratified squamous and stratified cuboidal; basal cells cuboidal or columnar; surface cells dome shaped or squamouslike, depending on degree of organ stretch.



Function: Stretches readily and permits distension of urinary organ by contained urine.

Location: Lines the ureters, bladder, and part of the urethra.



Transitional epithelium

Basement membrane
Connective tissue

Photomicrograph: Transitional epithelium lining of the bladder, relaxed state (500×); note the bulbous, or rounded, appearance of the cells at the surface; these cells flatten and become elongated when the bladder is filled with urine.

4-Transitional Epithelia

- **Locations**
- Transitional epithelium lines parts of the urinary system, where stretching is required:
- Urinary bladder
- Ureters
- Part of the urethra

4-Transitional Epithelia

- **Appearance (important for histology)**
- Empty bladder (relaxed):
- Surface cells appear rounded or dome-shaped
- Full bladder (distended):
- Surface cells appear flat
- Epithelium becomes thinner

4-Transitional Epithelia

- **Functions**
- Allows stretching (distension) of urinary organs
- Maintains an impermeable barrier to urine
- Protects underlying tissues from toxic urine contents

Transitional Epithelium



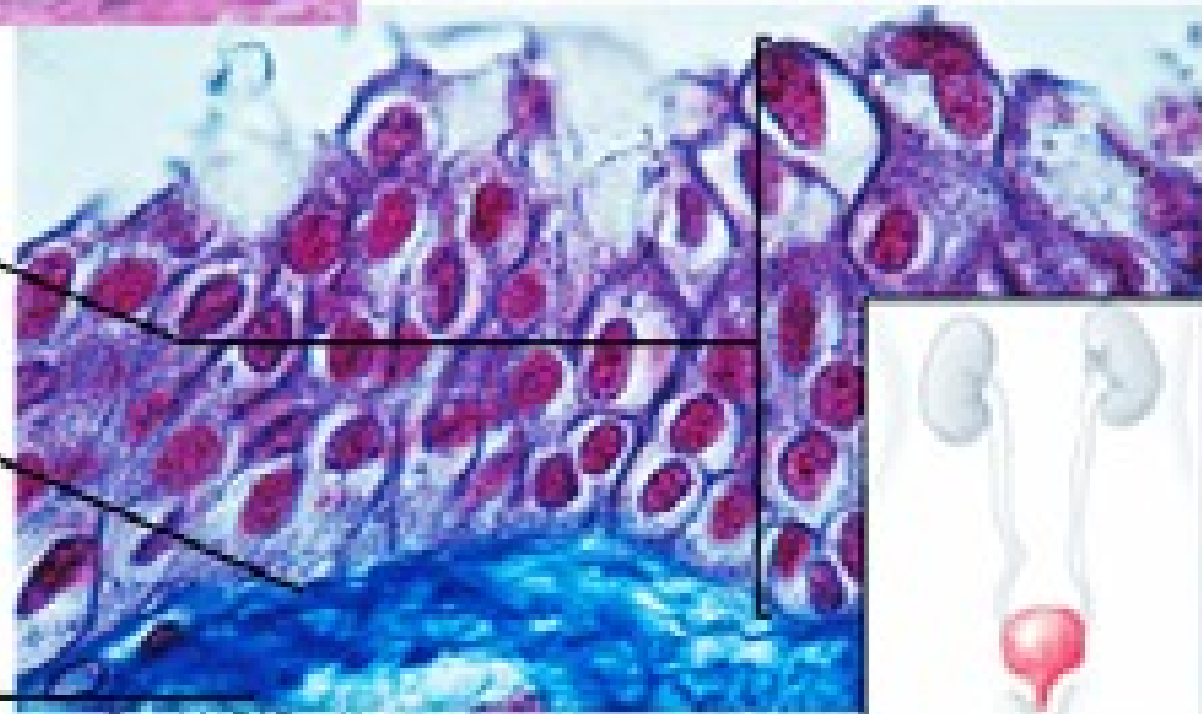
Basement
membrane

Underlying
connective
tissue

Unstretched
transitional
epithelium

Basement
membrane

Underlying
connective
tissue



MCQ:

- 1. Which epithelial tissue is best suited for diffusion and filtration?
 - A. Simple cuboidal epithelium
 - B. . Stratified squamous epithelium
 - c. . Simple squamous epithelium
 - D. . Transitional epithelium

- 2. An epithelium that appears stratified, but in which all cells rest on the basement membrane, is called:
 - A. Stratified columnar epithelium
 - B. Simple columnar epithelium
 - C. Transitional epithelium
 - D. Pseudostratified columnar epithelium

- 3. Which feature is most characteristic of transitional epithelium?
- A. Presence of microvilli
- B. Ability to stretch and change shape
- C. Single layer of flat cells
- D. Keratinized surface cells

- 4. Stratified squamous epithelium is named according to the shape of:
 - A. Basal cells
 - B. Middle layer cells
 - C. Surface (apical) cells
 - D. All cell layers

- 5. Which of the following locations is lined by simple cuboidal epithelium?
- A. Alveoli of lung
- B. Epidermis of skin
- C. Kidney tubules
- D. Urinary bladder.



Thanks